

Group Index, Ring Loss and Bend Loss of TM Silicon Strip Waveguides

edX: asafco91

Abstract

Design proposal for chip EBeam_asafco91. All circuits use the TM0 mode of a 500×220 nm Si strip waveguide. Two MZIs ($\Delta L = 100/200$ μm) give the group index (simulated $n_g = 3.765$ at 1550 nm). Five all-pass rings give the propagation loss. A bend cutback gives the loss per 5 μm bend.

1 Introduction

TM circuits need n_g , propagation loss and the minimum bend radius. TM modes are weakly confined, so bend loss is significant at small radii.

Objectives:

- $n_g(\lambda)$ of TM0 from the MZI FSR;
- propagation loss from the intrinsic Q of rings;
- loss per bend at $R = 5$ μm .

2 Theory

2.1 Waveguide Compact Model

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (1)$$

2.2 MZI Transfer Function

Two ideal 3 dB 2×2 couplers [1]:

$$\frac{I_{\text{bar}}}{I_i} = \frac{1}{4} \left| e^{-j\beta_1 L_1 - \frac{\alpha}{2} L_1} - e^{-j\beta_2 L_2 - \frac{\alpha}{2} L_2} \right|^2, \quad \beta = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (2)$$

Identical, lossless arms, $\Delta L = L_2 - L_1$:

$$\frac{I_{\text{bar, cross}}}{I_i} = \frac{1}{2} [1 \mp \cos(\beta \Delta L)]. \quad (3)$$

2.3 Group Index from the FSR

Arm phase difference $\phi = 2\pi n_{\text{eff}} \Delta L / \lambda$. Adjacent minima differ by $\Delta\phi = 2\pi$, so $|d\phi/d\lambda| \text{FSR} = 2\pi$ with

$$\frac{d\phi}{d\lambda} = -\frac{2\pi\Delta L}{\lambda^2} \left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right) \Rightarrow n_g = \frac{\lambda^2}{\Delta L \cdot \text{FSR}}. \quad (4)$$

For several ΔL :

$$\frac{1}{\text{FSR}} = \frac{n_g}{\lambda^2} \Delta L, \quad \frac{\delta n_g}{n_g} \approx \sqrt{2} \frac{\delta \lambda_{\text{min}}}{\text{FSR}}. \quad (5)$$

2.4 Ring Resonator

All-pass ring, $r = \sqrt{1 - \kappa^2}$, $a = e^{-\alpha L/2}$, $\theta = \beta L$ [2]:

$$T = \frac{a^2 - 2\text{arccos}\theta + r^2}{1 - 2\text{arccos}\theta + (ar)^2}, \quad Q_i = \frac{2\pi n_g}{\lambda \alpha}. \quad (6)$$

Critical coupling at $r = a$.

2.5 Bend Cutback

$$T_{\text{dB}}(N) = T_0 - N \text{IL}_b - \alpha_{\text{dB}} L(N). \quad (7)$$

3 Modelling and Simulation

3.1 Waveguide

Si strip, $w = 500$ nm, $h = 220$ nm, SiO_2 cladding, TM polarization (PDK type “Strip TM 1550 nm, w=500 nm” [3]). Lumerical MODE (FDE) [4]: Figs. 1, 2.

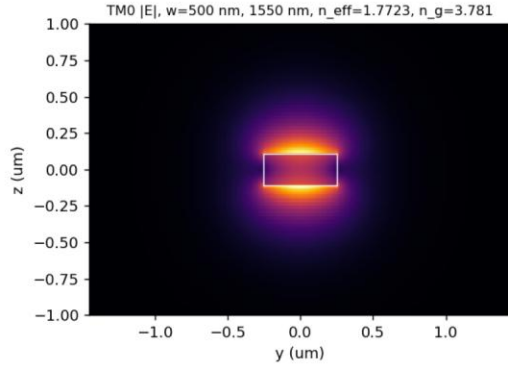


Fig. 1. TM_0 mode profile ($|E|$) at 1550 nm, MODE (FDE).

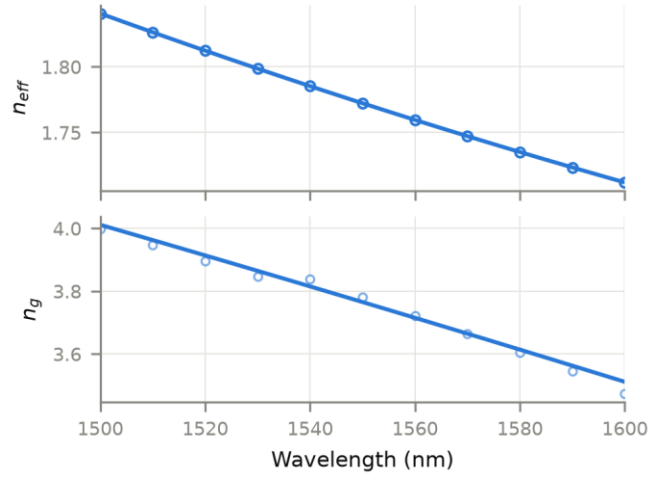


Fig. 2. n_{eff} and n_g vs. wavelength. Markers: MODE (FDE); lines: Eq. (8).

Compact model, λ in μm , $\lambda_0 = 1.55 \mu m$:

$$n_{eff}(\lambda) = 1.7723 - 1.2859(\lambda - \lambda_0) + 1.6131(\lambda - \lambda_0)^2. \quad (8)$$

$n_g = 4.011 / 3.765 / 3.511$ at 1500 / 1550 / 1600 nm: strong dispersion.

3.2 MZI Design

TABLE 1. Expected FSR from Eq. (8)

Device	ΔL (μm)	FSR (nm) at $\lambda =$			pts/ FSR
		1500	1550	1600	
MZITMw500dL100	100	5.61	6.38	7.29	5609
MZITMw500dL200	200	2.80	3.19	3.65	2805

ebeam_adiabatic_tm1550 split and combine; TM GCs. FSR < 50 nm and > 2800 points per FSR at 1 pm (Table 1).

3.3 Circuit Simulation

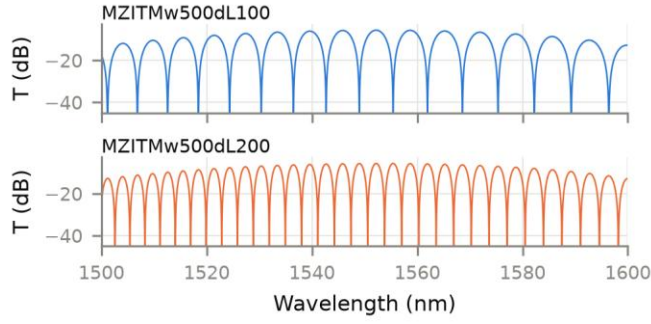


Fig. 3. Simulated MZI transmission (cross port), GC to GC: Eq. (3) with Eq. (8) and the PDK TM GC response [3].

Fig. 3.

3.4 Extraction Test on Simulated Spectra

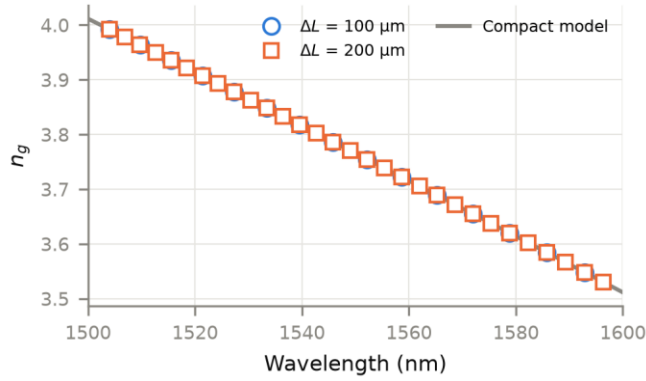


Fig. 4. n_g from the simulated FSR, Eq. (4) (markers), and Eq. (8) (line).

TABLE 2. n_g at 1550 nm extracted from Fig. 3

Device	FSR (nm)	n_g
MZITMw500dL100	6.381	3.7654
MZITMw500dL200	3.190	3.7654
Slope, Eq. (5), MZITMw500		3.7655

Eqs. (4), (5) recover the model $n_g(\lambda)$ (Fig. 4, Table 2).

3.5 Procedure for Measured Data

1. Subtract the LBTM spectrum (dB).
2. Locate the minima; quadratic fit around each null (sub-pm).
3. FSR of adjacent minima $\rightarrow n_g(\lambda)$, Eq. (4).

4. Slope over ΔL , Eq. (5); compare with Eq. (8).

3.6 Ring Resonators

TABLE 3. Expected ring response at 1550 nm ($R = 30 \mu\text{m}$, 3 dB/cm assumed)

Device	κ^2	FSR (nm)	Q_L	ER (dB)
RingTMg550	0.0733	3.72	30k	2.7
RingTMg600	0.0462	3.72	44k	4.4
RingTMg650	0.0290	3.72	63k	7.4
RingTMg700	0.0182	3.72	87k	13.3
RingTMg800	0.0071	3.72	138k	12.1

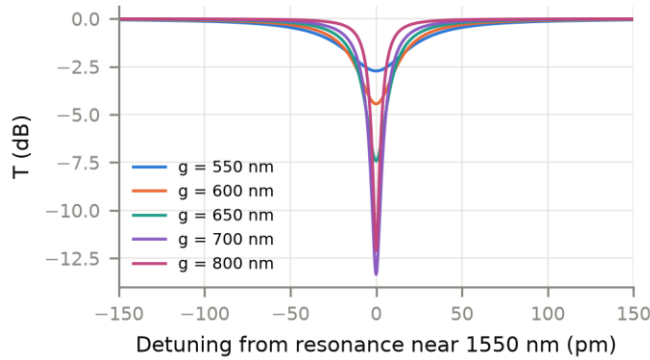


Fig. 5. Expected ring through-port spectra, Eq. (6).

$R = 30 \mu\text{m}$ (low TM bend loss), $L = 171.4 \mu\text{m}$. κ^2 from MODE supermode splitting of the coupler region [4]. Round-trip loss $1 - \alpha^2 = 0.0118$ vs. $\kappa^2 = 0.073\text{--}0.007$: the gaps span over- to under-coupling (Table 3, Fig. 5). Expected $Q_i \approx 221\text{k}$. Measured: fit Eq. (6) per resonance $\rightarrow r, a \rightarrow \alpha$; the gap series resolves the $r \leftrightarrow a$ ambiguity.

3.7 Bend Cutback

TABLE 4. Expected bend loss, $R = 5 \mu\text{m}$, U-turn serpentine

Device	N	L (μm)	$N \text{ IL}_b$ (dB) at $\lambda =$		
			1500	1550	1600
BendTMN0	0	175	0.0	0.0	0.0
BendTMN8	8	238	0.6	1.8	5.4
BendTMN16	16	300	1.3	3.5	10.7
BendTMN22	22	327	1.8	4.8	14.7

MODE bent-mode solver [4]: $\text{IL}_b \approx 0.22 \text{ dB per } 90^\circ$ at 1550 nm (radiation + two straight-bend junctions per U-turn; $\pm 50\%$), rising steeply with λ (Table 4). Measured: linear fit of Eq. (7) vs. N at each λ , after subtracting αL from the rings.

3.8 Loopback

$IL_{GC} = 1/2 T_{LB}$ [dB]. Expected (PDK TM GC model [3]): LBTM peak -5.6 dB at 1553 nm, 3 dB bandwidth 67 nm.

4 Fabrication

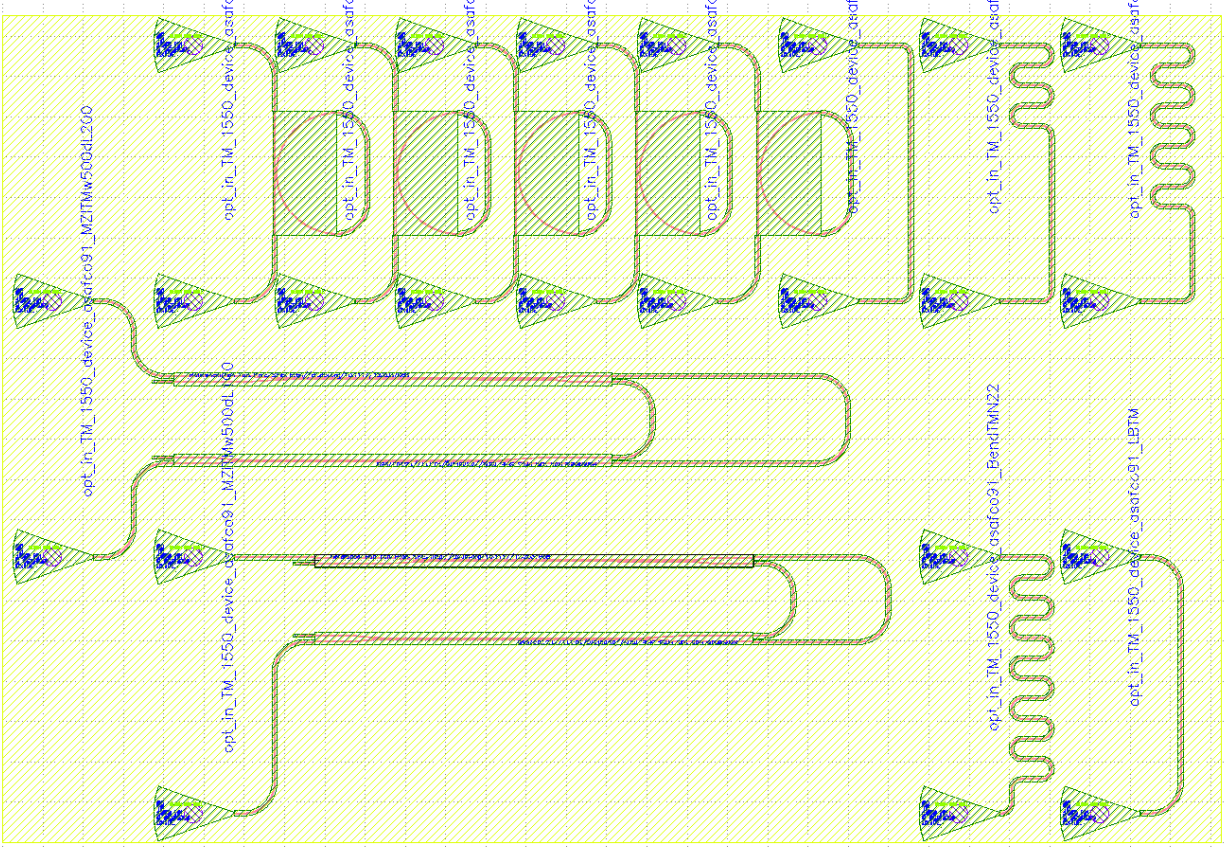


Fig. 6. Layout of EBeam_asafco91 ($605 \times 410 \mu\text{m}$).

TABLE 5. Devices on EBeam_asafco91

Device	Pol.	Description
MZITMw500dL200	TM	MZI, adiabatic 2×2 , $\Delta L = 200 \mu\text{m}$, $w = 500$ nm
MZITMw500dL100	TM	MZI, adiabatic 2×2 , $\Delta L = 100 \mu\text{m}$, $w = 500$ nm
RingTMg550	TM	all-pass ring, $R = 30 \mu\text{m}$, $g = 550$ nm
RingTMg600	TM	all-pass ring, $R = 30 \mu\text{m}$, $g = 600$ nm
RingTMg650	TM	all-pass ring, $R = 30 \mu\text{m}$, $g = 650$ nm
RingTMg700	TM	all-pass ring, $R = 30 \mu\text{m}$, $g = 700$ nm
RingTMg800	TM	all-pass ring, $R = 30 \mu\text{m}$, $g = 800$ nm
BendTMN0	TM	bend cutback, $N = 0$, $R = 5 \mu\text{m}$

Device	Pol.	Description
BendTMN8	TM	bend cutback, $N=8$, $R=5\text{ }\mu\text{m}$
BendTMN16	TM	bend cutback, $N=16$, $R=5\text{ }\mu\text{m}$
BendTMN22	TM	bend cutback, $N=22$, $R=5\text{ }\mu\text{m}$
LBTM	TM	GC loopback, $L=154\text{ }\mu\text{m}$

KLayout, SiEPIC-Tools [5], EBeam PDK [3] (Fig. 6, Table 5). TM GCs, $127\text{ }\mu\text{m}$ pitch; labels opt_in_TM_1550_device_asafco91_<name>. Functional check: 0 errors; DRC: 0 errors. Electron-beam lithography, Applied Nanotools NanoSOI [6], SiEPIC openEBL.

5 Experimental Data

To be completed (1500–1600 nm sweep, 1 pm step).

6 Analysis

To be completed.

7 Conclusion

To be completed.

References

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- [2] W. Bogaerts et al., “Silicon microring resonators,” *Laser & Photonics Reviews*, vol. 6, no. 1, pp. 47–73, 2012.
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